



Direct numerical simulation of liquid ammonia/methane dual-fuel swirl combustion under high pressure conditions

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ABSTRACT

Ammonia is an ideal zero-carbon fuel and it can be used directly in a liquid state. However, the utilization of liquid ammonia in practical combustors is constrained by its low reactivity. Direct injection of liquid ammonia co-firing with high-reactivity fuels is a promising solution, but its combustion characteristics are not well understood yet. In the present work, direct numerical simulation (DNS) of liquid ammonia/methane dual-fuel swirl combustion under high pressure conditions was performed in a Eulerian–Lagrangian framework to understand the flame characteristics and the complex turbulence–flame interactions. Two DNS cases with NH_3 energy ratios of 13.2% (Case L) and 38.72% (Case H) were considered. It was found that the flames are characterized with methane/air premixed combustion in the upstream region, and ammonia combustion dominates the downstream region. The flames are stabilized in the outer shear layer of the swirl flow with strong heat release rate (HRR) in the upstream region. No evident HRR is observed in the inner shear layer. The overall combustion intensity of case L is higher than that of case H, as expected. In the upstream region, the carbon-based reactions provide a significant contribution to the total HRR. As the axial distance increases, the nitrogen-based reactions make a substantial contribution to the total HRR where ammonia combustion dominates. Two additional DNS cases were carried out to understand the effects of droplet size and phase-change model on the structures of the swirling flame. It was shown that smaller droplets are more concentrated near the axis, where the flow velocity is larger and the fluid temperature is lower. When a classical evaporation model is used as the phase-change model, the evaporation rate is reduced and the droplets traverse further downstream, resulting in a longer flame.

1. Introduction

Ammonia (NH_3) is considered as a promising fuel due to its carbon-free nature, well-established production infrastructure, and efficient storage and transportation capabilities [1]. Therefore, the large-scale application of ammonia in power equipments can not only reduce the greenhouse effect but also alleviate the energy crisis from the combustion side. However, the utilization of ammonia suffers from the issues of ignition and flame stabilization due to its low reactivity compared to traditional hydrocarbon fuels [2]. Consequently, there is considerable interest in achieving effective ammonia combustion, which is vital in advancing zero-carbon energy utilization.

The feasibility of using ammonia in combustors has been extensively investigated. The swirl generates recirculating flows that promote transport of heat and active radicals, which is an effective technique to improve the stability and efficiency of combustion. In recent years, a number of studies on ammonia/air swirl combustion have been performed. Zhang et al. [3] experimentally investigated the blow-off

characteristics and macro structures of ammonia/air flames in a swirl combustor. Wei et al. [4] employed large-eddy simulation to understand flame topologies and the blow-off characteristics of ammonia/air swirl flames. Deng et al. [5] carried out numerical investigations of combustion dynamics in ammonia/air premixed swirl flames. These studies have demonstrated that ammonia/air flames could be successfully stabilized in a swirl combustor in a limited range of equivalence ratio. However, ammonia/air flames are featured with relatively weak combustion intensity and a longer combustion chamber is required for complete combustion compared to hydrocarbon/air combustors [6].

One of the most promising techniques to improve ammonia combustion performance is dual-fuel combustion of ammonia blended with high-reactivity fuels. Ji et al. [7] experimentally studied the structure of ammonia/methane co-firing premixed swirl flames. The experimental results showed that the addition of methane strengthens the combustion intensity due to the enhanced resistance to stretching. Kim et al. [8] experimentally investigated the combustion characteristics of

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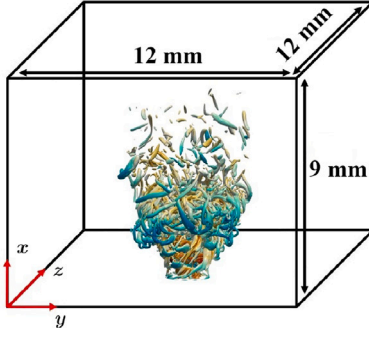


Fig. 1. Schematic of computational domain.

ammonia/hydrogen/air premixed flames in a swirl model combustor, and they observed that the combustion stability limits are extended with increasing volume fraction of H_2 in the fuel blend. Wang et al. [9] examined the combustion characteristics of partially premixed ammonia/methane dual swirl flames. The results show a central ammonia jet flame surrounded by compact swirl methane flames.

The above-mentioned studies focused on gaseous ammonia swirl combustion. In fact, ammonia can be used directly in a liquid state for energy systems. For gas turbines, supplying ammonia in liquid state to the combustor can get rid of the pre-vapor equipment, which reduces the cost and start-up time compared to supplying ammonia in gaseous phase. Okafor et al. [10] reported the first successful experimental study on liquid ammonia co-firing with methane in a gas turbine-like combustor. Uchida et al. [11] developed a technology to burn ammonia by directly injecting it into a combustor in a liquid state co-firing with natural gas. Recently, Wang et al. [12] experimentally investigates the topological structure of liquid ammonia spray swirl flames and discusses the influence of preheated air and hydrogen blending ratio on the flame topological structure. These studies provide valuable insights into the direct injection of liquid ammonia co-firing with high-reactivity fuels in a swirl combustor.

Direct numerical simulation (DNS) resolves all the flow and flame structures and plays an important role in revealing the combustion characteristics. However, high-fidelity numerical studies on direct injection of liquid ammonia co-firing with high-reactivity fuels are very scarce. Somaratne et al. [13] investigated liquid ammonia/hydrogen dual-fuel swirl combustion at moderately high pressure using large-eddy simulation. The numerical results showed the instantaneous temperature distributions at various dual-fuel blend ratios, but the underlying physics were not discussed. The combustion process of liquid ammonia combustion under gas turbine conditions is complex due to the interactions of turbulence, droplet and chemistry, which is difficult to be revealed using conventional numerical approaches. To our best knowledge, there have been no high-fidelity direct numerical simulations of turbulent liquid ammonia/methane dual-fuel swirl combustion under gas turbine conditions, which motivates the present work.

Based on the above discussion, the aim of the present work is to provide insights into the combustion characteristics of liquid ammonia/methane dual-fuel swirl flames with various NH_3 energy ratios, which is not well understood yet. The objectives are three-fold. First, three-dimensional DNS of turbulent liquid ammonia/methane dual-fuel swirl combustion using detailed chemical mechanism are performed under high pressure conditions. Second, the combustion characteristics of the flames are analyzed and the effects of NH_3 energy ratio on the flames are examined. Finally, the combustion mode of dual-fuel swirl flames is studied. The effects of droplet size and phase-change model on the flow and flame structures are also explored.

Table 1

Simulation parameters of various cases.

Case	ϕ_g	ER	ϕ_m	$\dot{m}_a$ (g/s)
L	0.8	13.2%	0.7	0.01969
H	0.8	38.72%	0.5	0.05918

2. Configuration and computational details

The schematic of the configuration is shown in Fig. 1. The swirl flow at the inlet consists of methane/air mixtures with a preheated temperature of 800 K. The pressure is 10 atm. These operating condition represents characteristic conditions for gas turbine applications [14]. The inlet is featured by a concentric annulus, with an inner diameter d of 0.5 mm and an outer diameter d_{out} of 1.7 mm. Liquid ammonia is injected into the computational domain from the central bluff-body. Lean combustion with a global equivalence ratio of $\phi_g = 0.8$ is considered. Two DNS cases with different NH_3 energy ratio (ER) are carried out, and the parameters of various cases are presented in Table 1. NH_3 energy ratio is defined as the ratio of the lower heating value (LHV) of NH_3 to that of the fuel mixture as:

$$ER = \frac{n_{NH_3} LHV_{NH_3}}{n_{NH_3} LHV_{NH_3} + n_{CH_4} LHV_{CH_4}} \quad (1)$$

where n_{CH_4} and n_{NH_3} are the mole fractions of methane and NH_3 , respectively. LHV_{CH_4} and LHV_{NH_3} are the lower heating values of methane and NH_3 , respectively. ϕ_m is the equivalence ratio of methane/air mixture, and $\dot{m}_a$ is the mass flow rate of liquid ammonia.

For both cases in this study, the swirl number is 1.0 and the Reynolds number is 5000. The liquid ammonia is assumed to be fully atomized, and the point-source method is employed for tracking ammonia droplets in a Eulerian-Lagrangian framework. The injection angle of liquid ammonia is 20° , and the Rosin-Rammler model is chosen for describing the droplet size distribution with a mean diameter of 8 μm . The droplet temperature is 279 K, and a similar droplet temperature was used in the experimental investigation of liquid ammonia spray co-firing with methane in a model combustor [10]. Note that the ammonia fuel is in a liquid state under the pressure of 10 atm. However, as ammonia droplets are heat by the hot ambient, flash boiling can take place. It is expected that variations in ambient temperature and pressure would result in different levels of flash boiling for ammonia droplets. In order to reasonably characterize the phase-change process of liquid ammonia, the flash boiling model developed by Zuo et al. [15] was employed.

The size of the computational domain is 9.0 mm in the streamwise direction, x , and is 12.0 mm in the lateral directions, y and z . Non-reflecting boundary conditions are applied to the inflow and outflow boundaries in the x direction, while no-slip adiabatic walls are imposed on the boundaries in the y and z directions. A hexahedral mesh system with three refinement levels is used. The resultant grid number for case L is $N_x \times N_y \times N_z = 330 \times 270 \times 270$, and that for case H is $N_x \times N_y \times N_z = 380 \times 290 \times 290$. The minimum grid size is 15 μm for both cases. The size of the Kolmogorov scale is about 9 μm , which is solved with more than 0.5 grid [16]. There are at least 10 grid points across the flame thickness with the grid resolution. Therefore, the flame and turbulence structures are properly resolved.

A detailed chemical mechanism for ammonia/methane combustion, which consists of 130 reactions and 42 species, is employed. This mechanism has been validated against experimental results of laminar burning velocity and Markstein lengths of ammonia/methane dual-fuel flames at high pressure conditions [17].

A DNS solver for spray combustion has been developed in the present work based on EBI-DNS, which has been validated extensively for turbulent combustion in the literature [18]. The original EBI-DNS solver was derived from the open-source software OpenFOAM [19]. The Lagrangian approach for sprays has been incorporated into the EBI-DNS

solver, and the equations for the droplets are solved [20]. The transport equations of continuity, momentum, energy and species mass fractions are solved for the gas phase in an Eulerian way, with the coupling terms between the gas and liquid phases being accounted for. The readers are referred to [20] for more details of the coupling between the gas and liquid phases. Cantera [21] is used to compute detailed chemistry and transport properties. The coupling of the velocity and pressure solution is based on the compressible PIMPLE algorithm. Time is discretized using an implicit second-order backward method. The cubic scheme based on a third-order polynomial fit for interpolation is employed for spatial discretization.

3. Results and discussion

3.1. General characteristics of swirl flames

To describe the general characteristics of liquid ammonia/methane dual-fuel swirl flames, Fig. 2 shows the instantaneous distributions of temperature, heat release rate (HRR) and mass fractions of H_2O and NO in a typical $x-r$ plane for both cases. The white dots represent ammonia droplets. The black dashed line outlines the inner shear layer (ISL) and outer shear layer (OSL). The location of the shear layer is defined as the location where the local velocity gradient is maximum. For both cases, the flames are characterized with methane combustion in the upstream region, and ammonia combustion dominates in the downstream region. It is observed that the gaseous temperature in the spray region drops sharply due to the large latent heat of liquid ammonia. Considering the higher flow rate of liquid ammonia in case H, the low-temperature region is much larger than that in case L. The flames are stabilized in the outer shear layer with strong HRR in the upstream region, while no evident HRR is observed in the inner shear layer due to the cooling effect from the evaporation of liquid ammonia, which is different from the scenario of gaseous ammonia/methane combustion [9,22,23], where both inner and outer flame branches were observed. It is readily seen that the overall combustion intensity of case L is higher than that of case H, as expected. There are pockets of NH_3 in the product region, which is further decomposed to radicals such as H and NH_2 , as shown the supplementary material. The peak mass fraction of H_2O in case L is lower than that in case H due to the higher flow rate of liquid ammonia in case H. Note that ammonia combustion generates a lot of H_2O . It can be observed that the distribution of NO corresponds to the high-temperature region. Therefore, there is also a significant difference in the distribution of NO mass fraction between the two cases. Particularly, high concentrations of NO are observed when combustion completes after $x/d = 6$ in case L while NO mainly appears in the hot product that is away from the jet center in case H. Fig. 2 also indicates that there are complex interactions between turbulence and scalar fields, so that the distributions of scalars are rather wrinkled, which are important features of turbulent combustion. These interactions are examined in terms of the alignment characteristics of the principal strain rates and scalar gradient in the supplementary material.

The statistical behavior of the flames is quantified next. The mean and standard deviation of HRR, and the mass fractions of H_2O and NO conditioned on mixture fraction at three axial locations are presented in Fig. 3. The mixture fraction (Z) in this study is consistent with Bilger's definition [24], which is defined based on the elemental mass fractions of C, H, O and N, as $Z = (\beta - \beta_a)/(\beta_f - \beta_a)$, where

$$\beta = 2 \frac{Y_C}{W_C} + 0.5 \frac{Y_H}{W_H} + 0.75 \frac{Y_N}{W_N} - \frac{Y_O}{W_O} \quad (2)$$

The subscripts 'f' and 'a' correspond to the fuel stream and the ambient, respectively. Y_i and W_i are the mass fraction and atomic weight of the i th element, respectively.

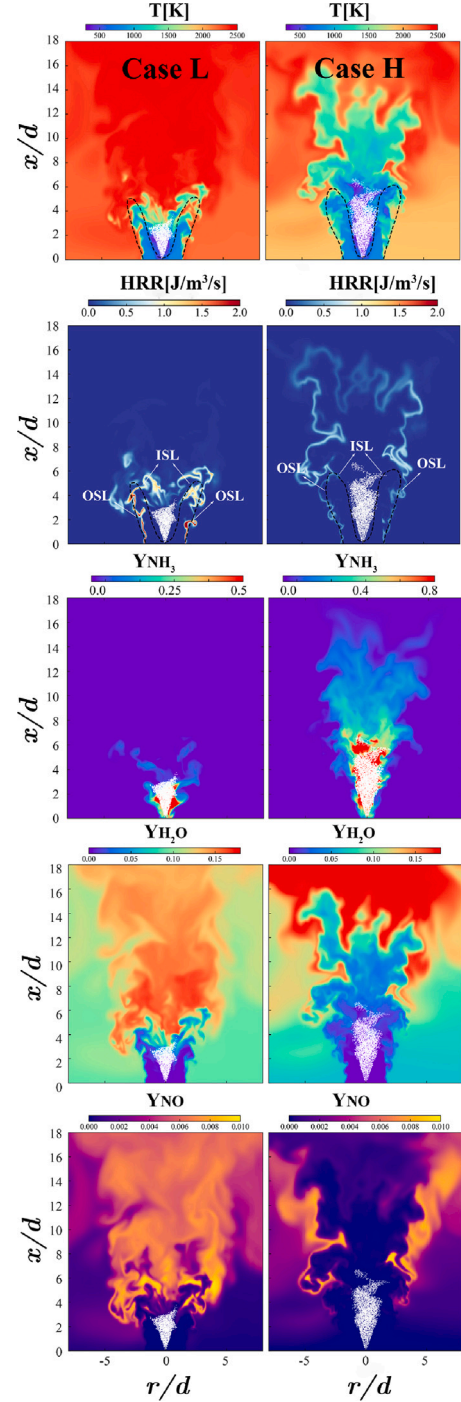


Fig. 2. Instantaneous contours of temperature, heat release rate and mass fractions of NH_3 , H_2O and NO in a typical $x-r$ plane for various cases. The white dots represent ammonia droplets. The black dashed line outlines the inner and outer shear layers.

It can be observed that at $x/d = 1$ for case L and $x/d = 2$ for case H, HRR is confined in narrow mixture fraction regions due to the fact that only methane/air premixed combustion occurs at these locations. HRR is distributed over a wider range of mixture fractions in the downstream region due to the evaporation of liquid ammonia, which increases the value of mixture fraction. The peak of H_2O and NO mass fractions in the two cases generally increases and shifts to richer mixtures as the axial distance increases.

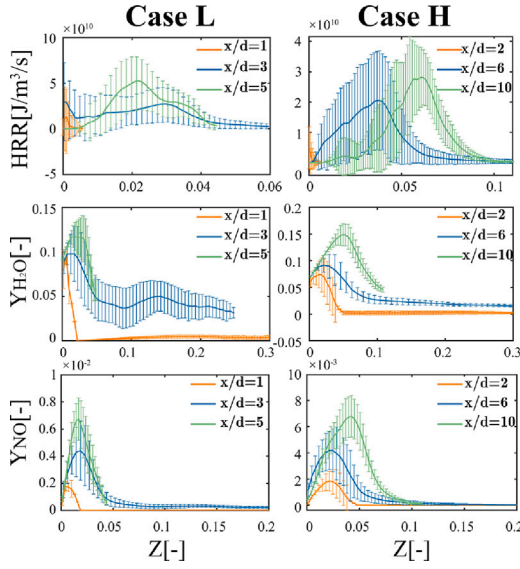


Fig. 3. The mean and standard deviation of heat release rate, and mass fractions of H₂O and NO conditioned on mixture fraction at three axial locations for various cases.

3.2. Combustion mode of dual-fuel flames

The evaporation of ammonia droplets not only leads to inhomogeneities in the fuel mass fraction field, but also results in the complex combustion modes characterized with premixed and non-premixed combustion. In order to identify the combustion modes of liquid ammonia/methane dual-fuel swirl flames, the normalized flame index (N.F.I.) proposed by Yamashita et al. [25] is used in the present work:

$$\text{N.F.I.} = n_{Y_F} \cdot n_{Y_{O_2}} = \frac{\nabla Y_F \cdot \nabla Y_{O_2}}{|\nabla Y_F| |\nabla Y_{O_2}|} \quad (3)$$

where Y_F and Y_{O_2} are the mass fractions of fuel and oxidizer, respectively. Here, $Y_F = Y_{CH_4} + Y_{NH_3}$. It is noted that N.F.I. is positive for premixed combustion and negative for non-premixed combustion. The instantaneous distributions of HRR in a typical $x-r$ plane for various cases are shown in Fig. 4. The black solid line represents the flame front, which is defined based on the progress variable calculated as $c = (Y - Y_{min}) / (Y_{max} - Y_{min})$. Here, $Y = Y_{H_2O} + 3.5Y_{CO_2}$ for case L and $Y = Y_{H_2O} + 2.5Y_{CO_2}$ for case H. Y_{max} and Y_{min} represent the maximum and minimum values of Y within the computational domain, respectively. The iso-surface of $c = 0.6$ is used to denote the flame front for both cases, which corresponds to the maximum heat release rate location. HRR from premixed (N.F.I. > 0) and non-premixed (N.F.I. < 0) combustion is denoted in red and blue, respectively. For both cases, the upstream region is featured by methane/air premixed combustion. Non-premixed combustion appears at the periphery of the reaction zone as the axial distance increases, where the combustion of ammonia becomes dominant. It is interesting to note that premixed combustion still prevails for ammonia combustion in the downstream region due to the fast evaporation process and the well mixing of the ammonia vapor and oxidizer.

Fig. 5 presents the instantaneous profiles of CH₄ and NH₃ reaction rates and HRR at three axial locations for various cases. The combustion modes are also marked in the plot. Particularly, red represents premixed combustion, and blue indicates non-premixed combustion. In the upstream region for the two cases, the reaction rate of ammonia is nearly zero. HRR is generated by methane combustion, which takes place in the premixed combustion zone. In the downstream region, the reaction rate of methane decreases, and that of ammonia increases. In the further downstream region for the two cases, the reaction rate of

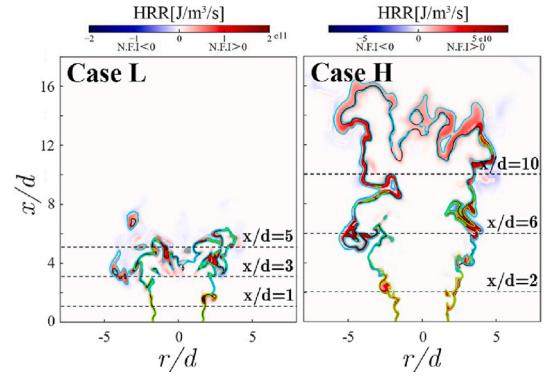


Fig. 4. Instantaneous contours of heat release rate in a typical $x-r$ plane for various cases. The heat release rate of premixed and non-premixed combustion is denoted by red and blue respectively. The chartreuse solid line represents the isoline of the 20% maximum methane reaction rate, while the cyan solid line represents the isoline of the 20% maximum ammonia reaction rate.

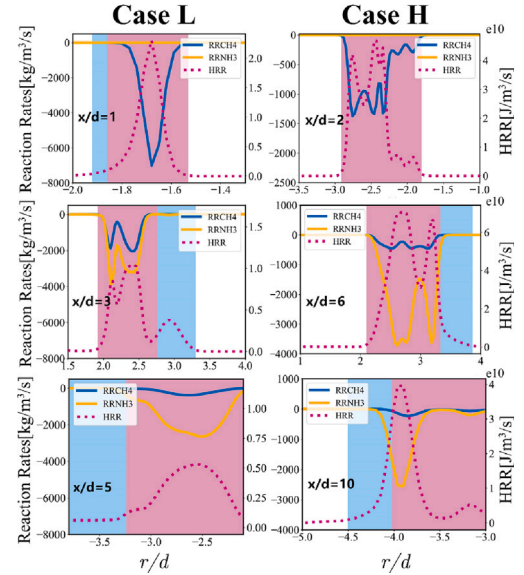


Fig. 5. Comparisons of CH₄ and NH₃ reaction rate and heat release rate at three axial locations for various cases. The background colors are used to distinguish the combustion modes: red represents premixed combustion and blue indicates non-premixed combustion.

ammonia dominates, while the reaction rate of methane is relatively low. It is interesting to see that the non-premixed combustion zone is featured by a relatively weak fuel reaction rate and heat release rate.

In order to quantify the roles of premixed and non-premixed combustion in the streamwise direction, a parameter W_p is introduced to characterize the contribution of premixed combustion to the total heat release of the gaseous combustion, which is calculated as:

$$W_p(x) = \frac{E_p}{E_t} = \frac{\iint \text{HRR}(x, y, z) |_{\text{N.F.I.} > 0} dy dz}{\iint \text{HRR}(x, y, z) dy dz} \quad (4)$$

where E_p is the integrated heat release of premixed combustion, and E_t is the total integrated heat release of both premixed and non-premixed combustion. Fig. 6 shows the profiles of W_p , E_p and E_t as a function of axial distance for cases L and H. As can be seen, the overall contribution of premixed combustion decreases with increasing axial locations. However, premixed combustion contributes more than non-premixed combustion to the total HRR even in the downstream region, for both cases. Note that the evaporation of ammonia droplets

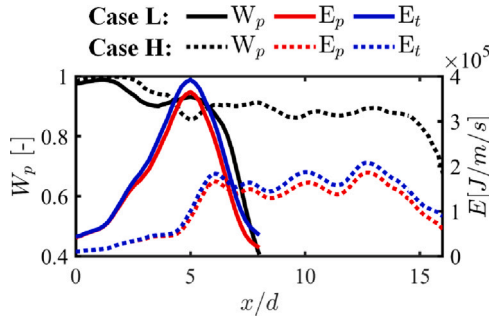


Fig. 6. The distributions of W_p , E_p and E_t as a function of the distances of streamwise direction for various cases.

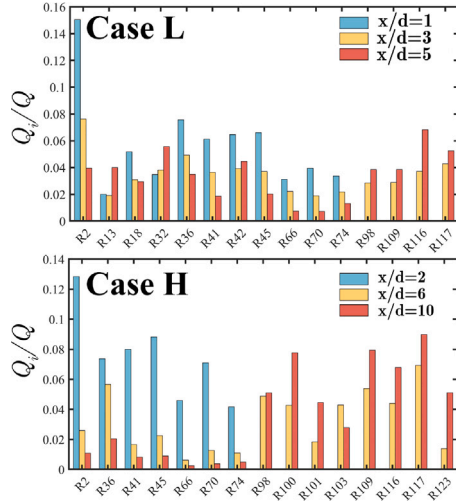


Fig. 7. Contributions of main exothermic reactions to total heat release rate at three axial locations for various cases.

completes in the upstream region (See Fig. 2). The gaseous ammonia then traverses downstream and mixes with the abundant oxidizer in the swirling flow, which results in the observed premixed combustion in the downstream region.

To further understand the combustion characteristics of the liquid ammonia/methane dual-fuel swirl flames, the contributions of main exothermic reactions to the total HRR at three axial locations for various cases are shown in Fig. 7. The relevant reactions, which locally contribute to more than 4% of the total heat release rate, are listed in Table 2. In the upstream region of both cases, HRR is generated by methane/air combustion. As can be seen, the contribution of $O + CH_3 = H + CH_2O$ is the largest, and several reactions related to OH consumption also provide a significant contribution. As the axial distance increases, the nitrogen-based reactions make a substantial contribution to the total HRR as ammonia combustion dominates the combustion process. Moreover, case H is featured by a larger number of nitrogen-based reactions with significant contributions to the total HRR compared to case L. This is due to the higher ammonia energy ratio in case H compared to case L.

3.3. Effects of droplet size and phase-change model

In the present work, the ammonia droplets are injected into a preheated mixture of methane/air, which can be superheated, and flash boiling occurs. It is interesting to use a non-flash boiling model, and compare the predictions with different phase-change models. Moreover, in practical applications, very small ammonia droplets can be generated due to flash boiling. Therefore, the case with smaller droplets

Table 2

Main reactions that contribute to the total HRR.

	Reaction
R2:	$O + CH_3 = H + CH_2O$
R13:	$H + OH = H_2O$
R18:	$H + CH_3 = CH_4$
R32:	$OH + H_2 = H + H_2O$
R36:	$OH + HO_2 = H_2O + O_2$
R41:	$OH + CH_4 = CH_3 + H_2O$
R42:	$CO + OH = CO_2 + H$
R45:	$OH + CH_2O = HCO + H_2O$
R66:	$2CH_3 = C_2H_6$
R70:	$HCO + O_2 = HO_2 + CO$
R74:	$CH_2 + O_2 = 2H + CO_2$
R98:	$NH_2 + OH = NH + H_2O$
R100:	$NH_2 + NH_2 = NH_3 + NH$
R101:	$NH_2 + NH = N_2H_2 + H$
R103:	$NH_2 + NO = N_2 + H_2O$
R109:	$NNH + O_2 = N_2 + HO_2$
R116:	$NH_3 = NH_2 + H$
R117:	$NH_3 + OH = NH_2 + H_2O$
R123:	$N_2H_2 + H = NNH + H_2$

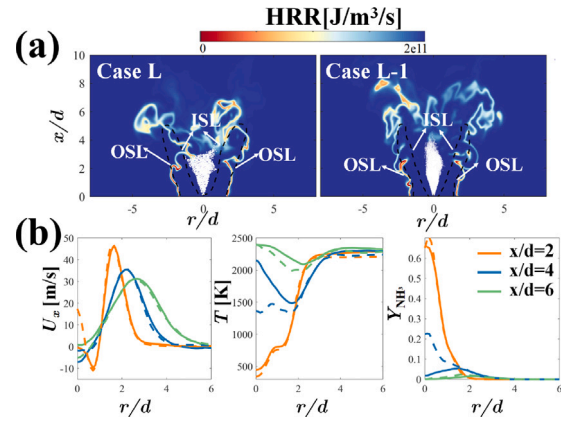


Fig. 8. (a) Instantaneous contours of heat release rate with the white dots representing ammonia droplets. (b) Profiles of mean axial velocity, temperature and NH_3 mass fraction for (solid line) case L and (dashed line) case L-1.

should also be investigated. In this subsection, two additional DNS cases, *i.e.* case L-1 and case L-2, are carried out to understand the effects of droplet size and phase-change model on the flow and flame structures. The numerical setup of the two cases is consistent with that of case L. However, in case L-1 the mean droplet diameter is reduced to 5 μm , while in case L-2 a classical equilibrium model [26] is employed as the phase-change model.

The instantaneous distributions of heat release rate and droplets for cases L and L-1 are shown in Fig. 8a. It is clear that the droplets are more concentrated near the axis in case L-1 compared with case L. This is because the evaporation of droplets near the edge of the spray cone completes faster in case L-1 with smaller droplets. The mean profiles of axial velocity, temperature and NH_3 mass fraction are compared between the two cases in Fig. 8b. The main difference of the mean fields between the two cases is observed near the axis. The large number of droplets near the axis in case L-1 lead to an increased flow velocity and reduced temperature compared to case L.

The effect of phase-change model on the structures of the swirling flame is examined in Fig. 9, which shows the instantaneous distributions of heat release rate and droplets for cases L and L-2, and compares the mean profiles of axial velocity, temperature and NH_3 mass fraction are compared between the two cases. As seen, the droplets without flash boiling appear further downstream. Therefore, the fluid temperature is generally lower in case L-2 compared with case L. More gaseous

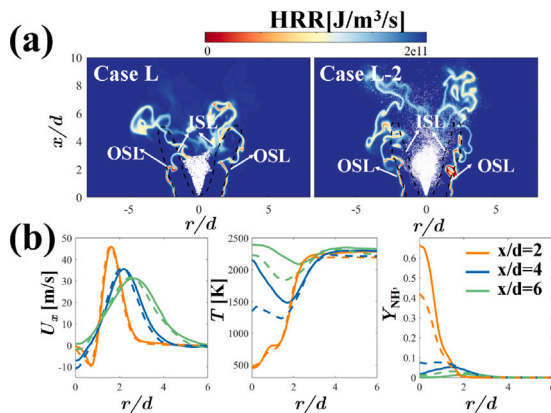


Fig. 9. (a) Instantaneous contours of heat release rate with the white dots representing ammonia droplets. (b) Profiles of mean axial velocity, temperature and NH_3 mass fraction for (solid line) case L and (dashed line) case L-2.

ammonia is available for chemical reactions in the downstream region of case L-2, resulting in a longer flame.

4. Conclusions

In the present work, the first DNS of liquid ammonia/methane dual-fuel swirl combustion was performed to investigate the flame characteristics with varying NH_3 energy ratio. The flames are characterized with methane combustion in the upstream region, and ammonia combustion dominates in the downstream region. The overall combustion intensity of case L is higher than that of case H, as expected. HRR is distributed over a wider range of mixture fractions in the downstream region due to the evaporation of liquid ammonia, which increases the value of mixture fraction. The peak of H_2O and NO mass fractions in the two cases generally increases and shifts to richer mixtures as the axial distance increases.

The complex combustion modes are studied. It was found that the overall contribution of premixed combustion decreases with increasing axial locations. However, premixed combustion still contributes more than non-premixed combustion to the total HRR even in the downstream region for both cases. It was observed that the contributions of $\text{O} + \text{CH}_3 = \text{H} + \text{CH}_2\text{O}$ is the largest in the upstream region, and several reactions related to OH consumption also provide a significant contribution. As the axial distance increases, nitrogen-based reactions make a substantial contribution to the total HRR as ammonia dominates in the combustion process.

Two additional DNS cases were carried out to understand the effects of droplet size and phase-change model on the flow and flame structures. It was shown that, in the case with a smaller droplet size, the droplets are more concentrated near the axis, where the flow velocity is larger and the fluid temperature is lower. When a classical evaporation model is used as the phase-change model, the evaporation rate is reduced and the droplets traverse further downstream in the computational domain, resulting in a longer flame. This work presents a detailed analysis of liquid ammonia/methane swirl combustion relevant to gas turbine applications, which complements the previously reported experimental studies, and provides insights into the spray and combustion characteristics under various operating conditions. However, in reality the atomization process might also be influenced by flash boiling. Our future work will consider the atomization of liquid ammonia in the simulations.

Novelty and significance statement

The paper reports the first direct numerical simulation of liquid ammonia/methane dual-fuel swirl combustion under high pressure conditions. The novelty of this research is that it reveals the complex combustion characteristics and turbulence-flame interactions of dual-fuel flames with various NH_3 energy ratios, which is not well understood yet. The instantaneous combustion characteristics of various flames are analyzed and the complex combustion modes are studied. The effects of droplet size and phase-change model on the flow and flame structures are also examined. The study is significant as it improves our fundamental understanding of liquid ammonia/methane dual-fuel swirl combustion characteristics, which is crucial for the development advanced gas turbine with zero-carbon fuels.

CRediT authorship contribution statement

Ziwei Huang: Performed research, Analyzed data, wrote the paper. **Haiyou Wang:** Supervised the project, Designed research, Provided fundings, Reviewed the paper. **Qian Meng:** Analyzed data, Wrote the paper. **Kun Luo:** Reviewed the paper. **Jianren Fan:** Supervised the project, Reviewed the paper.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.proci.2025.105962>.

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